Project Workflow Manager

Software Requirements Specification

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1. Introduction

1.1 Purpose

The purpose of this document is to outline the complete software requirements, including system architecture and functionality, and to guide project development moving forward. This Software Requirements Specification (SRS) provides a comprehensive description of the system and contains the information necessary for software engineers to design and implement the product. The intended audience for this document includes the software development team, where it serves as a blueprint and guideline during development, as well as stakeholders who require a clear understanding of the system and its components.

1.2 Scope

The scope of this project includes two software products: a calendar-based project management application and a database designed to support it.
The calendar manager will contain a dashboard interface which will allow all users to view assignments and projects which they have been assigned to work on. Users will be able to see assignments, projects, and project phases they are assigned to, as well as the steps and due dates tied to them. Users will be able to create assignments and projects via the application. Project managers will be able to modify and archive projects and phases to include removal of assigned team members and the modification of steps. The system supports multi-user collaboration through role-based permissions.
The database serves as back-end persistent storage for the application. It will store items such as user account information, project information, association of users to projects, and personal responsibilities.
The goal of these applications is to provide centralized project visibility, calendar-based task tracking, structured phase management, and role-controlled collaboration to improve coordination. It is not designed to replace external storage or submission platforms such as GitHub or Brightspace.

1.3 Definitions, Acronyms, and Abbreviations

User: A user is anyone who utilizes the application.
Project Manager: A project manager is a user who creates, manages, and assigns Projects and Assignments to other users.
Assignment: A singular task or set of tasks assigned to one or more users.
Project: A larger Assignment which can be broken down into one or more smaller sub-assignments
Project Phase: A sub-assignment which makes up a large Project.
Project Steps: The task or tasks which make up a Project Phase.

1.4 Overview

The remainder of this document is comprised of four main sections. The next section provides a general description and information regarding the project, such as product functionality and user types. Section 3 details software development guidelines, with functional requirements and use cases being used to dictate development to ensure the final product functions as intended. Section 4 contains analysis models and sequence diagrams for the application. The final section details the guidelines and requirements for future modifications to this document.

2. General Description

2.1 Product Perspective

This product is a new, self-contained product separate from any existing line of applications currently available. While similar products such as Google Calendar and Brightspace do exist, the Project Workflow Manager does not mimic these products. Rather, it incorporates ideas and systems relative to these applications, and frames project management in a dependency-related manner, to provide something new.

2.2 Product Functions

- Creation of assignments, projects, and project phases by instructors/project managers
- Modification of information assigned to projects and assignments
- Viewing of assigned projects and assignments by users
- Creation and authentication of user accounts
- Assignment of users to projects and project phases
- Archiving projects and prevention of further modification
- Calendar-based visualization of assignments, projects, and dependencies
- Role-based permission control for project management actions

2.3 Users and Characteristics

There are two primary classes of users: Project Managers and Standard Users.
Project managers are users who oversee, manage, and assign work to other users. This may include an instructor, project lead, or administrator. Project managers can create, populate, and modify assignments, projects, and phases, as well as assign users to projects and phases. Project Manager privileges operate under authenticated, role-based permissions.

Standard Users are individuals assigned to projects, phases, or assignments by a Project Manager, such as students, team members, or contractors. Standard Users can view assignments, projects, and project phases to which they are assigned. Standard Users may create personal assignments but cannot modify or delete projects they do not manage.

2.4 General Constraints

- Time: This project has a strict time limit of one semester (four months)
- Money: This project does not have any financial backing from the university, meaning any costs for development must come from the developers' personal funds.
- Experience/knowledge: The development team is limited to three members, meaning only the knowledge currently held by team members and any knowledge which can be learned during the project's development time will be available for this project.
- Development resources (hardware, software, ext.): The resources for application development are limited to what is available for free, or for use via university lease and subscription.

2.5 Assumptions and Dependencies

- Assumption: The software will operate in an environment compatible with Windows 11. Changes to the supported operating system environment may require modification of system requirements.
- Assumption: The users of the application will have reliable internet connectivity when accessing the system. If network availability is inconsistent, functionality and requirements related to data access may need to be adjusted.
- Assumption: The users possess basic computer literacy and can navigate a web app. Significant differences in user technical abilities may require modifications to usability and interface requirements.
- Dependency: The system is dependent upon the availability of the database for maintaining pertinent system data. If the database is unavailable, data management may need revision.
- Dependency: The system is dependent on an authentication mechanism to verify user identity and enforce permissions. The system may require adjustments to security and access-control requirements.
- Dependency: The system is dependent on a calendar-based interface for visualization and organization of assignments and projects.

2.6 Operating Environment

The Project Workflow Manager will operate on laptop and desktop computer systems running a compatible operating system, including Windows 11. The system is intended to function within a standard user environment and will exist with common components like networking services and browser software. No specialized external applications are required.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

3.1.2 Hardware Interfaces

3.1.3 Software Interfaces

3.1.4 Communications Interfaces

3.2 Functional Requirements

3.2.1 Creating a Profile

3.2.1.1 Description and Priority

Any user can create a personal account on the system. This profile is used to authenticate users and to associate users with projects. A user with an account can log in and view projects, assignments, and other responsibilities assigned to them. This feature is a high priority.

3.2.1.2 Stimulus/Response Sequences

Stimulus: User selects “Create Account”
Response: System presents the account creation form.
Stimulus: User enters valid values for username, password, and company role.
Response: System validates that fields are valid against formatting rules.
Response: System checks to see if the requested username is already in use.

Response: System creates a new user record in persistent storage.
Response: System initializes the account with default permissions.
Response: System confirms successful account creation.
Response: System redirects user to login page.

3.2.1.3 Functional Requirements

REQ-1: The system shall provide an account creation interface accessible to users.
REQ-2: The system shall require a username, password, and company role during account creation.
REQ-3: The system shall validate account input fields against required formatting rules.
REQ-4: The system shall verify that the requested username is unique.
REQ-5: The system shall create and store a new user record upon successful registration.
REQ-6: The system shall initialize the account with default permissions.
REQ-7: The system shall notify the user if account creation fails due to invalid input or username.
REQ-8: The system shall redirect the user to the login interface upon successful account creation.

3.2.2 Adding an Assignment

3.2.2.1 Description and Priority

Any user that has passed authentication and can view their dashboard can add an assignment. An assignment is assigned a name and is associated with a due date. This assignment is displayed on the user's calendar. This is a low priority.

3.2.2.2 Stimulus/Response Sequences

Stimulus: User selects "Add Assignment"
Response: System presents the assignment creation form
Stimulus: User enters assignment/responsibility name and valid due date and clicks "Submit"
Response: System validates the assignment input fields against formatting rules
Response: System creates an assignment record in persistent storage associated with current user
Response: System displays confirmation of the creation of the assignment
Response: System redirects the user to the dashboard

3.2.2.3 Functional Requirements

REQ-1: The system shall allow authenticated users to create assignments.
REQ-2: The system shall require an assignment name and due date.
REQ-3: The system shall validate assignment input fields before creation.
REQ-4: The system shall associate the created assignment with the authenticated user.
REQ-5: The system shall store the assignment in persistent storage.
REQ-6: The system shall display the assignment on the user's calendar.
REQ-7: The system shall notify the user if assignment creation fails due to invalid input.

3.2.3 Create a Project

3.2.3.1 Description and Priority

A user that is authenticated and has permission to create a project can create a project. The creator of the project is automatically assigned to the project manager role and is visible within the creator's project list. This is a high priority.

3.2.3.2 Stimulus/Response Sequences

Stimulus: User selects "Create Project"
Response: System presents the project creation form

Stimulus: User enters fields for project name, due date, and associated steps
Stimulus: User selects "Submit"
Response: System validates the formatting of the field inputs
Response: System checks persistent storage for existing project name
Response: System creates a new project record in persistent storage.
Response: System displays a confirmation message that the project was created successfully.
Response: System redirects the user to the dashboard.

3.2.3.3 Functional Requirements

REQ-1: The system shall allow authenticated users with permissions to create projects.
REQ-2: The system shall require a project name and due date.
REQ-3: The system shall validate project input fields before creation.
REQ-4: The system shall verify that the project name is unique.
REQ-5: The system shall create and store the project in persistent storage.
REQ-6: The system shall assign the project creator the Project Manager role.
REQ-7: The system shall display the project on the dashboard calendar.
REQ-8: The system shall notify the user if project creation fails due to invalid input or project name.

3.2.4 Add Project Phases

3.2.4.1 Description and Priority

A project manager can add additional phases to a project. Multiple phases can occur during the same timeframe, split among teams. For example, Project 1 can be split into backend development and UI design phases to occur at the same time. The created phase can be assigned as a dependency to inform team members of due dates, responsibilities, and other parts of the project dependent on their completion. This feature is a medium priority.

3.2.4.2 Stimulus/Response Sequences

Stimulus: Project manager selects "View Projects".
Response: System retrieves and displays the list of projects associated with the user.
Stimulus: Project manager selects the desired project.
Response: System retrieves and displays the details for the selected project.
Stimulus: Project manager selects "Add Phase".
Response: System displays the phase creation form to include due date, name, and dependencies.
Stimulus: Project manager enters valid input into form fields and clicks "Submit".
Response: System validates fields and verifies unique phase name.
Response: System creates a new phase record associated with the project.
Response: System adds the project phase to the calendar display.
Response: System displays confirmation of phase creation.
Response: System redirects the project manager to the dashboard.

3.2.4.3 Functional Requirements

REQ-1: The system shall allow authorized users to add phases to a project.
REQ-2: The system shall require a phase name and due date.
REQ-3: The system shall validate phase input fields before creation.
REQ-4: The system shall verify that the phase name is unique within the project.
REQ-5: The system shall associate the phase with the selected project.
REQ-6: The system shall allow phases to be assigned dependencies.
REQ-7: The system shall store the phase in persistent storage.

REQ-8: The system shall display the phase on the project calendar.
REQ-9: The system shall notify the user if phase creation fails due to invalid input.

3.2.5 Assign Members to Project Phases

3.2.5.1 Description and Priority

A project manager can assign other users to project phases. This allows project managers to delegate tasks and responsibilities. The assigned users can view all details of their assigned project phase, including due date, requirements, and dependencies. This feature is a high priority.

3.2.5.2 Stimulus/Response Sequences

Stimulus: Project manager selects the desired project phase.
Response: System presents the project phase interface.
Stimulus: Project manager selects "Assign Team".
Response: System retrieves list of users from persistent storage.
Stimulus: Project manager selects users to add to the phase and clicks "Confirm".
Response: System allocates permissions to view the phase to selected users.
Response: System displays confirmation that permissions have been allocated.
Stimulus: User selects "Done".
Response: System redirects the project manager to the dashboard.

3.2.5.3 Functional Requirements

REQ-1: The system shall allow authorized users to assign members to project phases.
REQ-2: The system shall retrieve and display available users for selection.
REQ-3: The system shall associate selected users with the specified project phase.
REQ-4: The system shall grant assigned users permission to view the phase.
REQ-5: The system shall update assigned users' dashboards and calendars to reflect the assignment.
REQ-6: The system shall notify the user if assignment fails.

3.2.6 View Project Details

3.2.6.1 Description and Priority

Any user can view the details of a phase or project they are assigned to. This includes steps involved, due date, dependencies, and other pertinent information. The user can view the project node itself on the calendar but visit the project details for a full breakdown of the phase or project. This is a medium priority.

3.2.6.2 Stimulus/Response Sequences

Stimulus: User clicks on project node.
Response: System retrieves the project data and displays it on the project interface.

3.2.6.3 Functional Requirements

REQ-1: The system shall allow authorized users to view project or phase details.
REQ-2: The system shall retrieve project data from persistent storage.
REQ-3: The system shall display project name, due date, phases, dependencies, and assigned users.
REQ-4: The system shall prevent unauthorized access to project details.

3.2.7 View Project List

3.2.7.1 Description and Priority

Any user can view an organized list of the projects they are assigned to other than the calendar view. This allows for easier access to different project steps and project modification in a centralized location. This is a low priority.

3.2.7.2 Stimulus/Response Sequences

Stimulus: User selects "View Project List"
Response: System retrieves list of associated projects.
Response: System displays a list of associated projects ordered by due date.

3.2.7.3 Functional Requirements

REQ-1: The system shall allow authenticated users to view their associated projects.
REQ-2: The system shall retrieve projects associated with the user from persistent storage.
REQ-3: The system shall display projects ordered by due date.

3.2.8 Archive Project

3.2.8.1 Description and Priority

A project manager can archive a project. Archiving removes the project from active calendars and prevents further modification. The project remains visible in the project manager's project list for reference. This is a low priority.

3.2.8.2 Stimulus/Response Sequences

Stimulus: Project manager clicks "View Project List"
Response: System retrieves and displays the list of all projects associated with the manager.
Stimulus: Project manager selects the project they wish to archive.
Response: System displays the project viewing interface.
Stimulus: Project manager selects "Archive Project"
Response: System authenticates the project manager to confirm the archive.
Stimulus: Project manager selects "Confirm Archive"
Response: System disables active assignments and dependencies associated with the project.
Response: System notates project as "Archived" in project details.
Response: System redirects project manager to the project list interface.

3.2.8.3 Functional Requirements

REQ-1: The system shall allow authorized users to archive projects.
REQ-2: The system shall update the project status to "Archived".
REQ-3: The system shall remove archived projects from active calendar displays.
REQ-4: The system shall prevent modification of archived projects.
REQ-5: The system shall retain archived project data in persistent storage.

3.2.9 Modify Project

3.2.9.1 Description and Priority

A project manager may modify aspects of projects/phases. This includes project steps, due dates, assigned users, etc. A project manager must authenticate themselves before modification. These modifications are then updated for all affected users. This is a medium priority.

3.2.9.2 Stimulus/Response Sequences

Stimulus: Project manager selects the desired project node to modify.
Response: System displays project interface for specified project.
Stimulus: Project manager selects "Modify Project"
Response: System prompts authentication
Stimulus: User provides valid username and password
Response: System displays the modification interface.
Stimulus: Project manager unassigns members or changes steps or due date and clicks "Submit".
Response: System validates modifications.
Response: System updates project information in persistent storage.
Response: System displays confirmation of modification.
Response: System redirects to the previous interface.

3.2.9.3 Functional Requirements

REQ-1: The system shall allow authorized users to modify project information.
REQ-2: The system shall allow modification of project steps, due date, and assigned users.
REQ-3: The system shall validate modifications before saving.
REQ-4: The system shall update project information in persistent storage.
REQ-5: The system shall propagate modifications to affected users.

3.2.10 User Login

3.2.10.1 Description and Priority

A user with an existing account must authenticate themselves to access their dashboard. The user must enter their username and associated password at the login interface to gain access. If the information is incorrect, the user will not be permitted access to the application. This is a high priority.

3.2.10.2 Stimulus/Response Sequences

Stimulus: User enters valid username and password in the login interface fields and clicks "Login".
Response: System verifies password against username's associated password in storage.
Response: System redirects to the dashboard associated with the authenticated user account.

3.2.10.3 Functional Requirements

REQ-1: The system shall provide a login interface for users.
REQ-2: The system shall authenticate user credentials against stored account data.
REQ-3: The system shall deny access if authentication fails.
REQ-4: The system shall initiate a user session upon successful authentication.
REQ-5: The system shall redirect authenticated users to their dashboard.

3.3 Use Cases

3.3.1 Use Case #1

Use Case Name	Create Account
Reference	Section 3.2.1
Trigger	The user selects "Create Account" from the login interface.
Precondition	The system is operational. The user is on the login page. The user is not authenticated.
Basic Path	1. The user selects "Create Account" from the login interface.

	2. The system presents the registration interface with fields for username, password, and role. 3. The user enters username, password, and role preferences and selects "Submit." 4. The system validates the registration input against formatting rules. 5. The system queries persistent storage by username to verify that the username is not associated with another account. 6. The system creates and stores a new account record in persistent storage. 7. The system presents a message that confirms that account creation has been successful. 8. The system redirects the user to the login page to log in with their new credentials.
Alternative Paths	In step 4, if the input is invalid, the system presents a notification that the input was invalid. 4. The system detects that the registration input is invalid. 5. The system displays an input error notification. 6. The user clicks "Okay." Return to step 3. In step 5, if the username exists in persistent storage, the system presents a notification that the username is taken. 5. The system detects that the username already exists in persistent storage. 6. The system displays a username error notification. 7. The user clicks "Okay." Return to step 3.
Postcondition	A new, unique account record exists in persistent storage. The user is at the login screen.
Exception Paths	The user may abandon or cancel the account creation process at any time prior to submission. No account is created, and the use case terminates.
Other	(none)

Table 1: Use case 1

3.3.2 Use Case #2

Use Case Name	User Login
Reference	Section 3.2.10
Trigger	The user opens the application.
Precondition	The system is operational and accessible. The user is not currently authenticated.
Basic Path	1. The user opens the application. 2. The system presents the login page. 3. Include: *AuthenticateUser* (UC-03) 4. The system redirects the user to the dashboard associated with the authenticated account.
Alternative Paths	In step 3, if authentication fails: 3. Authentication fails within *AuthenticateUser* (UC-03) 4. The system remains on the login page. 5. Return to Step 2.
Postcondition	The user is authenticated. The user is viewing the dashboard associated with the authenticated account.

Exception Paths	The user closes the application before authentication completes. The use case terminates.
Other	

Table 2: Use case 2

3.3.3 Use Case #3

Use Case Name	AuthenticateUser (sub-use case)
Reference	Sections 3.2.8, 3.2.9, 3.2.10
Trigger	The system requests authentication, and the user submits their username and password.
Precondition	The system is operational. The user is not currently authenticated for the requested protected action.
Basic Path	1. System prompts the user to enter their username and password. 2. The user enters a valid username and password and selects "Submit". 3. System queries persistent storage for the username. 4. System verifies that the provided password matches the stored credentials for the account. 5. System confirms successful authentication.
Alternative Paths	In step 3, if the system cannot find the username in persistent storage: 3. The system cannot find the username in persistent storage. 4. The system displays a notification that the username cannot be found. 5. Return to Step 2. In step 4, if the system cannot match the provided password to the stored credentials: 4. The system cannot match the provided password to the stored credentials. 5. The system displays a notification that incorrect credentials were provided. 6. Return to Step 2.
Postcondition	The user is authenticated for the requested action (or session is confirmed)
Exception Paths	The user cancels authentication. The protected action is not performed, and the use case terminates.
Other	(none)

Table 3: Use case 3

3.3.4 Use Case #4

Use Case Name	Add Assignment
Reference	Section 3.2.2
Trigger	User selects "Add Assignment".
Precondition	User is authenticated. User is on the dashboard.
Basic Path	1. User selects "Add Assignment". 2. System presents the assignment creation interface. 3. User enters an assignment name and due date and selects "Submit". 4. System validates the user's input format. 5. System creates an assignment record associated with the user and the specified due date. 6. System adds an assignment representation on the calendar on the specified date. 7. System displays confirmation that assignment creation was successful. 8. System redirects the user to the dashboard.

Alternative Paths	At step 4, if the format of the input is invalid: 4. The system detects that the assignment input is invalid 5. The system displays an input error notification. 6. The user selects "Okay". 7. Resume at Step 3.
Postcondition	A new assignment record exists in persistent storage associated with the user. The assignment is visible on the user's calendar.
Exception Paths	The user cancels the assignment creation process prior to submission. No assignment is created, and the use case terminates.
Other	(none)

Table 4: Use case 4

3.3.5 Use Case #5

Use Case Name	Create Project
Reference	Section 3.2.3
Trigger	User selects "Create Project".
Precondition	User is authenticated. User is on the dashboard.
Basic Path	1. User selects "Create Project". 2. System presents the project creation interface. 3. User enters valid values for project name, due date, and project steps and selects "Submit". 4. System validates user input values against formatting rules. 5. System verifies that the project name does not already exist in persistent storage. 6. System creates a project record in persistent storage. 7. System assigns the user the project manager role for the created project. 8. System adds the project to the user's dashboard calendar. 9. System provides confirmation that the project was created successfully. 10. System redirects the user to the dashboard.
Alternative Paths	In step 4, if the input is invalid: 4. The system detects that one or more input fields are invalid. 5. The system displays an input error notification. 6. The user selects "Okay". 7. Return to Step 3. In step 5, if the project name exists: 5. The system detects that the project name already exists. 6. The system displays a duplicate name error notification. 7. The user selects "Okay". 8. Return to Step 3.
Postcondition	A new project record exists in persistent storage. The user is assigned the Project Manager role for the project. The project is visible on the dashboard calendar.
Exception Paths	The user cancels the project creation process prior to submission. No project is created, and the use case terminates.
Other	(none)

Table 5: Use case 5

3.3.6 Use Case #6

Use Case Name	Assign Workers to Project Phase
Reference	Section 3.2.5
Trigger	Project manager selects "Assign workers" in the phase interface.
Precondition	User is authenticated. User is the project manager or has project manager permissions. The project exists in persistent storage. At least one other user exists in persistent storage.
Basic Path	1. Project manager selects "Assign workers" in the phase interface. 2. System retrieves a list of existing user accounts. 3. System displays the worker assignment interface. 4. Project manager selects workers to assign and selects "Submit". 5. System creates record(s) associating each selected user with the selected project phase. 6. System adds the project phase node to the assigned users' calendar displays. 7. System displays confirmation that worker assignment was successful. 8. System redirects the project manager to the dashboard.
Alternative Paths	In step 4, if no workers are selected: 4. The system detects that no workers were selected. 5. The system displays a notification indicating that at least one worker must be selected. 6. Return to Step 3.
Postcondition	Selected users are associated with the project phase. The project phase appears on assigned users' calendars.
Exception Paths	The project manager cancels the assignment process before submission. No worker assignments are created, and the use case terminates.
Other	(none)

Table 6: Use case 6

3.3.7 Use Case #7

Use Case Name	Add Project Phases
Reference	Section 3.2.4
Trigger	Project manager selects "Add Project Phase" from the project details interface.
Precondition	User is authenticated. User has the project manager role for the project. The project exists in persistent storage.
Basic Path	1. Project manager selects "Add Project Phase" from the project details interface. 2. System displays the phase creation interface. 3. Project manager enters valid values for phase name, due date, and phase steps and selects "Submit". 4. System validates input against formatting rules. 5. System creates a record for the phase in storage and associates it with the project. 6. System adds the phase node to the project manager's calendar display. 7. System displays confirmation that the phase was created successfully.

	8. System redirects the project manager to the project details interface for the created phase.
Alternative Paths	In step 4, if the input is invalid: 4. The system detects that one or more input fields are invalid. 5. The system displays an input error notification. 6. Return to Step 3. In step 5, if the phase name already exists: 5. The system detects that a phase with the same name already exists for the project. 6. The system displays a duplicate name error notification. 7. Return to Step 3.
Postcondition	A new phase record exists in persistent storage associated with the project. The phase is visible in the project details and calendar display.
Exception Paths	The project manager cancels the phase creation process prior to submission. No phase is created, and the use case terminates.
Other	

Table 7: Use case 7

3.3.8 Use Case #8

Use Case Name	View Project Details
Reference	Section 3.2.6
Trigger	User selects a project node from the project interface or from the dashboard calendar.
Precondition	User is authenticated. User is on the dashboard or project list interface. The project exists in persistent storage. The user has permission to view the project.
Basic Path	1. User selects a project node from the calendar or project list. 2. System retrieves project data from persistent storage. 3. System displays the project details in the project details interface.
Alternative Paths	(none)
Postcondition	Project details are displayed to the user. No project data is modified.
Exception Paths	The project is no longer available, or the user loses authorization. The system displays an error message and returns the user to the previous interface.
Other	(none)

Table 8: Use case 8

3.3.9 Use Case #9

Use Case Name	View Project List
Reference	Section 3.2.7
Trigger	The user selects “Project List”.
Precondition	The user is authenticated. The user is assigned to or is the project manager of at least one project.
Basic Path	1. The user selects “Project List”. 2. System retrieves all project records associated with the user from persistent storage.

	3. System displays the list of projects ordered by due date.
Alternative Paths	In step 2, if the user has no associated projects: 2. The system determines that no projects are associated with the user. 3. The system displays an empty project list and a notification that no projects are available.
Postcondition	The project list is displayed to the user. No project data is modified.
Exception Paths	The system cannot retrieve project data. The system displays an error message and returns the user to the previous interface.
Other	(none)

Table 9: Use case 9

3.3.10 Use Case #10

Use Case Name	Archive Project
Reference	Section 3.2.8
Trigger	Project manager selects "Archive Project".
Precondition	User is authenticated. User is the project manager or has project manager permissions. The project exists in persistent storage. The project has not already been archived. The project is visible on their calendar and project list.
Basic Path	1. Project manager selects "Archive Project". 2. Include: *AuthenticateUser* (UC-03) 3. System updates the project status to "Archived" in persistent storage. 4. System removes the project from active calendar displays. 5. System prevents further modification of the archived project. 6. System displays confirmation that the project was archived successfully. 7. System redirects user to project list.
Alternative Paths	In step 2, if the user was unable to be authenticated: 2. Authentication fails within *AuthenticateUser* (UC-03). 3. The archive operation is not performed. 4. The system returns to the project details interface.
Postcondition	The project status is set to "Archived". The project is no longer visible in active calendar displays. The project remains stored for reference and cannot be modified.
Exception Paths	The project manager cancels the archive confirmation. No changes are made, and the use case terminates.
Other	(none)

Table 90: Use case 10

3.3.11 Use Case #11

Use Case Name	Modify Project
Reference	Section 3.2.9
Trigger	Project manager selects "Modify Project".
Precondition	User is authenticated.

	User is the project manager or has project manager permissions. The project exists in persistent storage. The project is visible on the calendar and in the project list.
Basic Path	1. Project manager selects "Modify Project" from the project details interface. 2. Include: *AuthenticateUser* (UC-03) 3. System displays the project modification interface. 4. Project manager enters valid updated values for due date and project steps and selects "Submit". 5. System validates that input against formatting requirements. 6. System updates the project record in persistent storage with the provided values. 7. System displays confirmation that the changes were successful. 8. System redirects the project manager to the project details interface.
Alternative Paths	In step 2, if authentication fails: 2. Authentication fails within *AuthenticateUser* (UC-03). 3. The modification operation is not performed. 4. The system returns to the project details interface. In step 5, if the input is invalid: 5. The system detects that one or more input fields are invalid. 6. The system displays an input error notification. 7. Return to Step 4.
Postcondition	The project record reflects the updated values. The updated project information is visible in the project details interface and calendar.
Exception Paths	The project manager cancels the modification process before submission. No changes are saved, and the use case terminates.
Other	(none)

Table 101: Use case 11

3.3.12 Use Case #12

Use Case Name	Add Project Dependencies
Reference	Section 3.2.4
Trigger	Project manager selects "Create Dependency" in the project details interface.
Precondition	User is authenticated. User is the project manager or has project manager permissions. The project exists in persistent storage. The project contains at least two phases.
Basic Path	1. Project manager selects "Create Dependency" in the project details interface. 2. System retrieves and displays a list of phases associated with the project. 3. Project manager selects a source phase and a dependent phase and selects "Submit". 4. System validates the dependency selection. 5. System stores a record in persistent storage representing the dependency relationship between the selected phases. 6. System updates the calendar display to visually represent the dependency between the phases. 7. System displays confirmation that the dependency was created successfully. 8. System redirects the project manager to the project details interface.

Alternative Paths	In step 4, if the dependency selection is invalid: 4. The system detects that the dependency selection is invalid (no phase selected or self-dependency). 5. The system displays an error notification. 6. Return to Step 3.
Postcondition	A dependency relationship exists in persistent storage between the selected phases. The dependency is visually represented in the project calendar display.
Exception Paths	The project manager cancels the dependency creation process before submission. No dependency is created, and the use case terminates.
Other	(none)

Table 112: Use case 12

3.4 Non-Functional Requirements

3.4.1 Performance

95% of user-initiated processes shall be completed and reflected in the user interface within 2 seconds under normal operating conditions.

3.4.2 Reliability

The system shall maintain a minimum uptime of 99% excluding scheduled maintenance.
In the event of a system failure, service shall be restored within 5 minutes.
No committed processes shall be lost in the event of a system crash.

3.4.3 Availability

The system shall maintain 99% availability annually, excluding scheduled maintenance.

3.4.4 Security

All users shall authenticate using unique credentials prior to being granted system access.
Access to projects and phases shall be restricted based on role-based permissions.
Unauthorized users shall not be permitted to view or modify protected data.
All stored credentials shall be protected using secure storage practices.

3.4.5 Maintainability

The system shall be developed using modular design principles to allow independent modification of functional components.
System documentation shall be maintained and updated with each major revision.
Database schemas shall adhere to standard normalization practices to ensure data integrity and maintainability.

3.4.6 Portability

The system is designed for desktop-class operating systems and is not required to support mobile operating systems.

3.5 Design Constraints

- The system shall require user authentication via username and password in accordance with project security requirements.

- The system shall use a persistent database for storage of user, project, assignment, and dependency data.
- The system shall operate on desktop and laptop systems running Windows 11.
- The system shall implement role-based access control to distinguish between project managers and standard users.
- The system shall maintain data storage such that user and project data is not lost between sessions.
- The database design shall enforce normalization and data integrity.
- The system shall be developed using tools that are freely available.
- The system shall support concurrent access by multiple users.

3.6 Logical Database Requirements

A logical database will be used to store user data and assignment/project data. Logical requirements for this database will include:

- User account records, each with their own unique identifier and securely stored credentials.
- Project/assignment records, each with their own unique identifier.
- Related records associated with projects, including phases, dependencies, and users.

The database will need to support at least 500 users, and at least 2000 projects/assignments. The database will need to retain this information for at least one year. Additionally, the database shall enforce data integrity through normalization, unique identifiers, and referential integrity between related records. This ensures consistency and support for future system expansion.

4. Analysis Models

4.1 Sequence Diagrams

4.1.1 Use Case 1

The following diagram illustrates the interactions between the user, system, and storage during the account creation process. The process begins with the user selecting to create an account. The system presents the registration form, and the user submits their username, password, and company role. The system validates the input and checks the database to ensure the username is unique. Once verified, the system stores the new account record in the database. After successful creation, the system redirects the user to the login page.

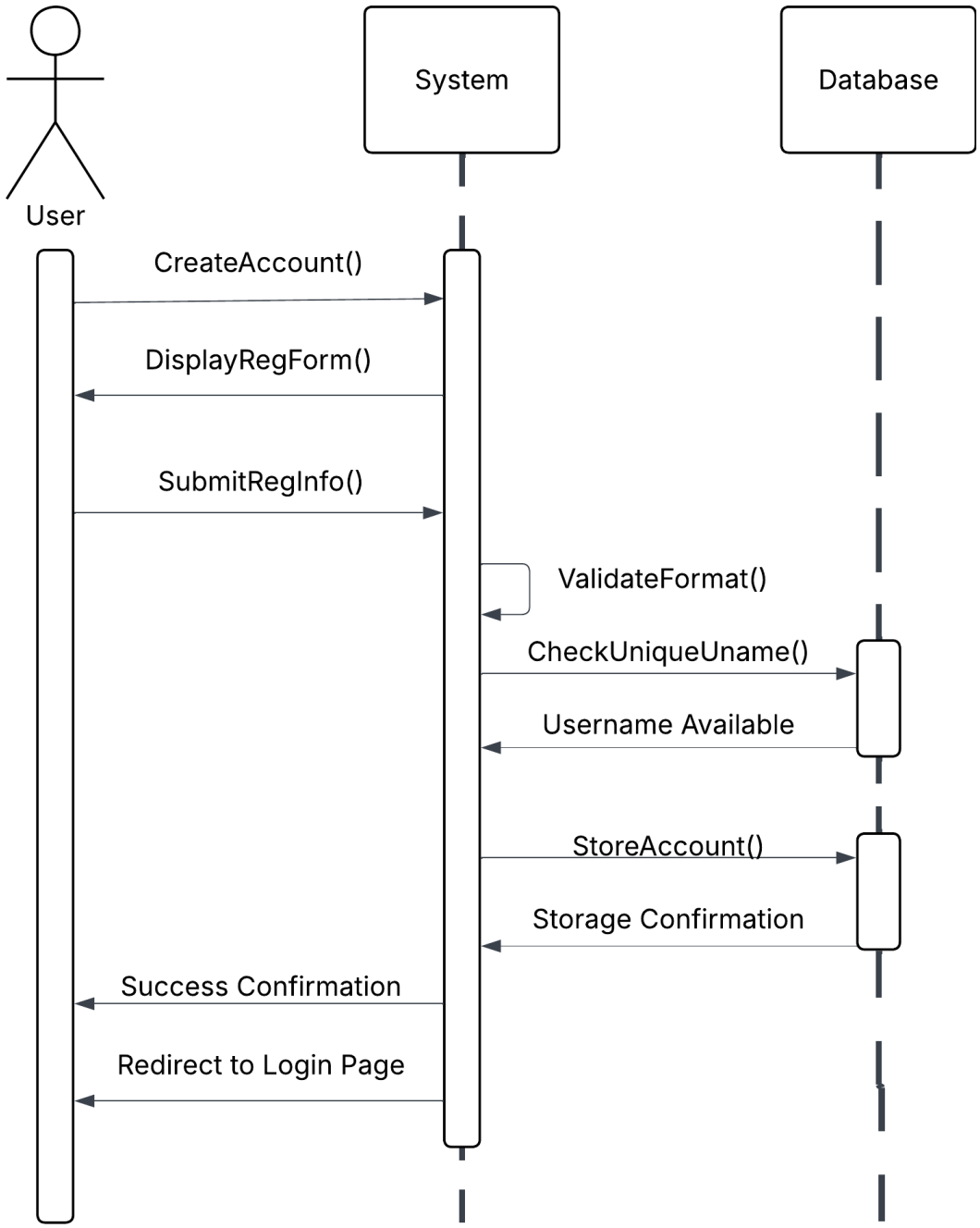

Figure 1: Data Flow Diagram Use Case 1

4.1.2 Use Case 2

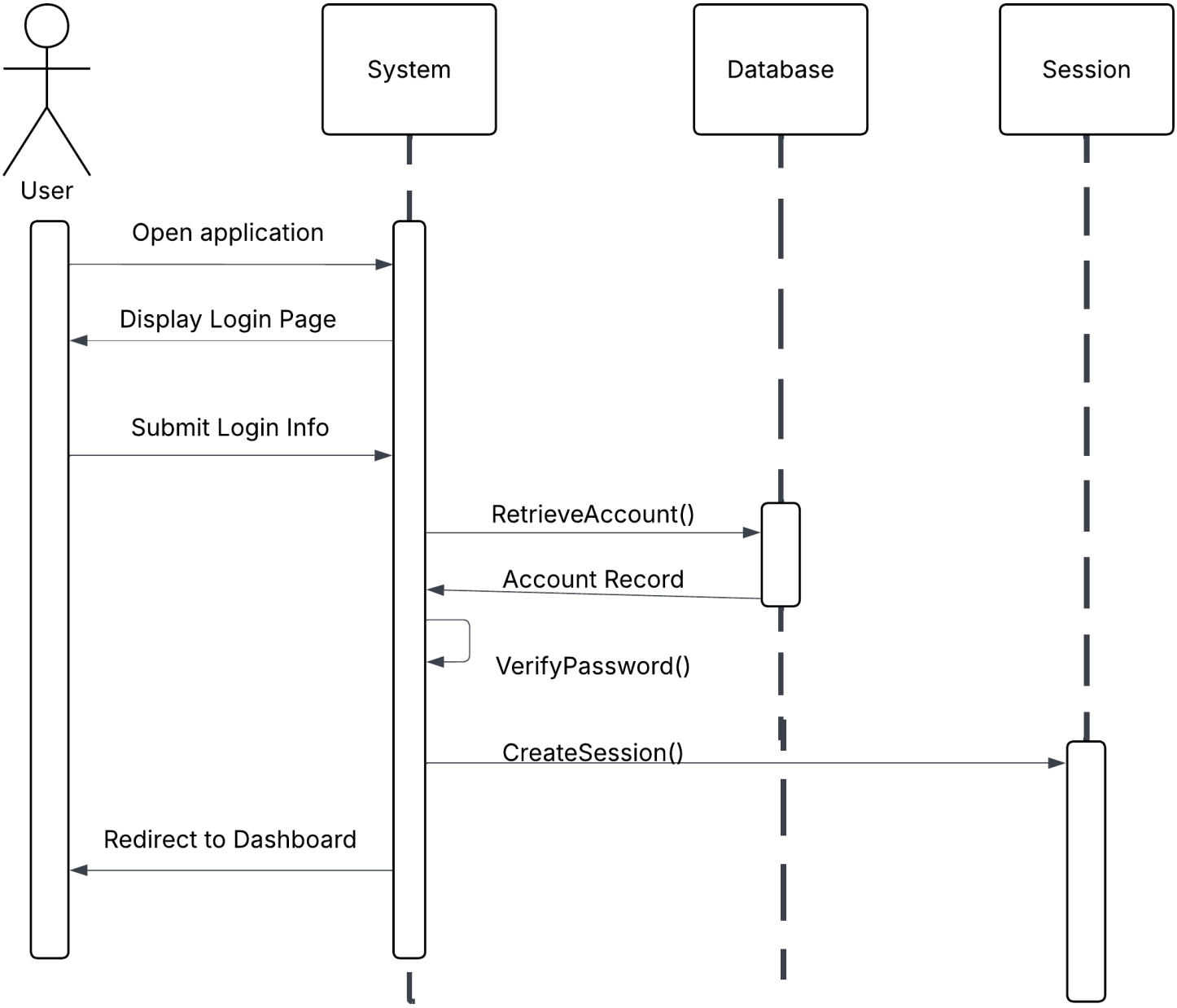

Figure 2 Data Flow Diagram Use Case 2

4.1.3 Use Case 3

The following diagram illustrates the authentication verification process used by protected system actions. The authentication process begins when the system prompts the user to enter their credentials. The user submits their username and password, and the system checks the database for the account record. The system verifies the password against stored credentials and confirms successful authentication. This then allows the requested action to proceed.

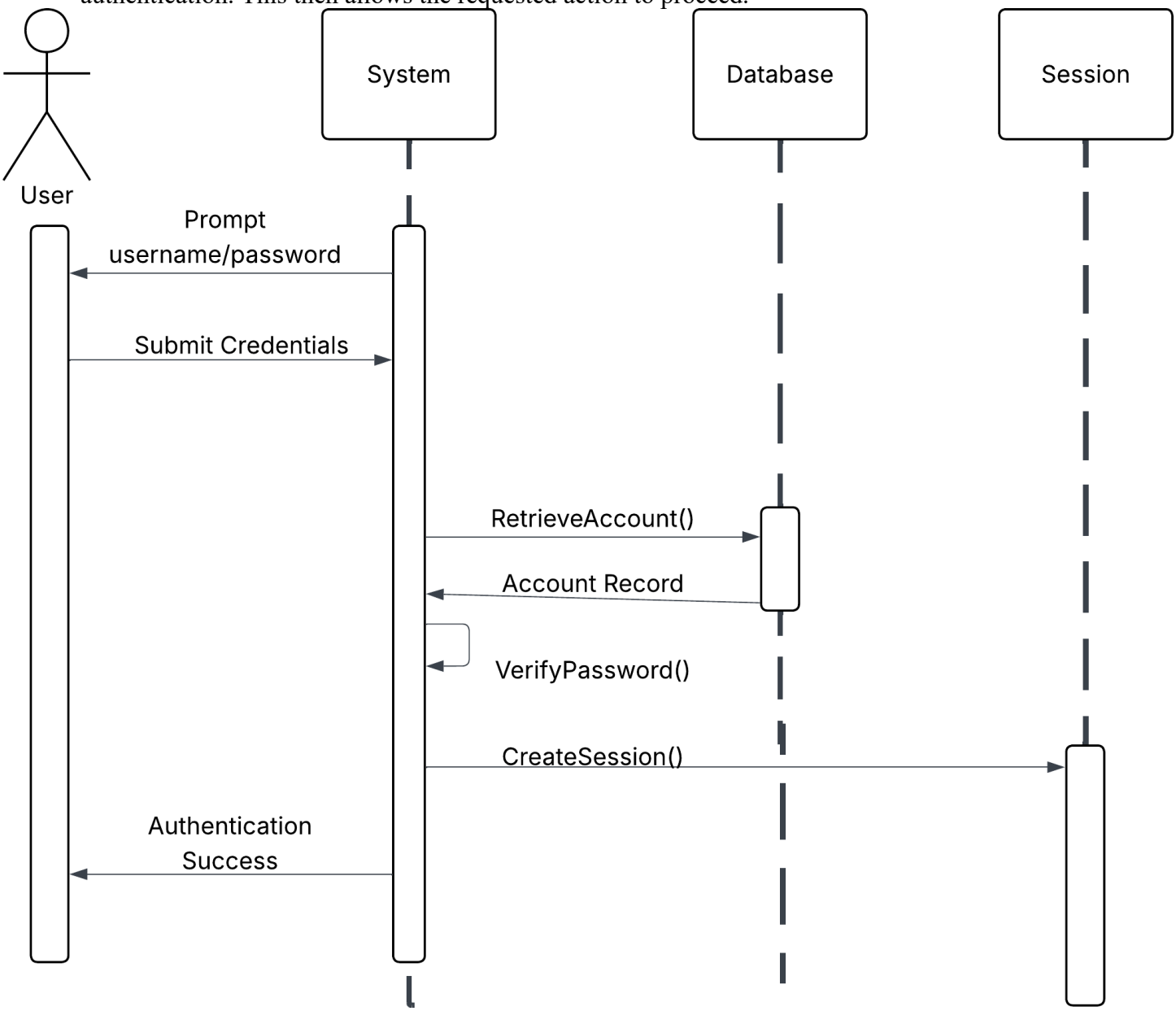

Figure 3 Data Flow Diagram Use Case 3

4.1.4 Use Case 4

The following diagram illustrates how a user creates a new assignment and how the system records and displays it. The process begins when the user selects the option to add an assignment. The system presents the assignment creation interface, and the user submits the assignment name and due date. The system validates the input and stores the assignment in the database, associated with the user. Once stored, the system updates the calendar display and confirms the successful creation.

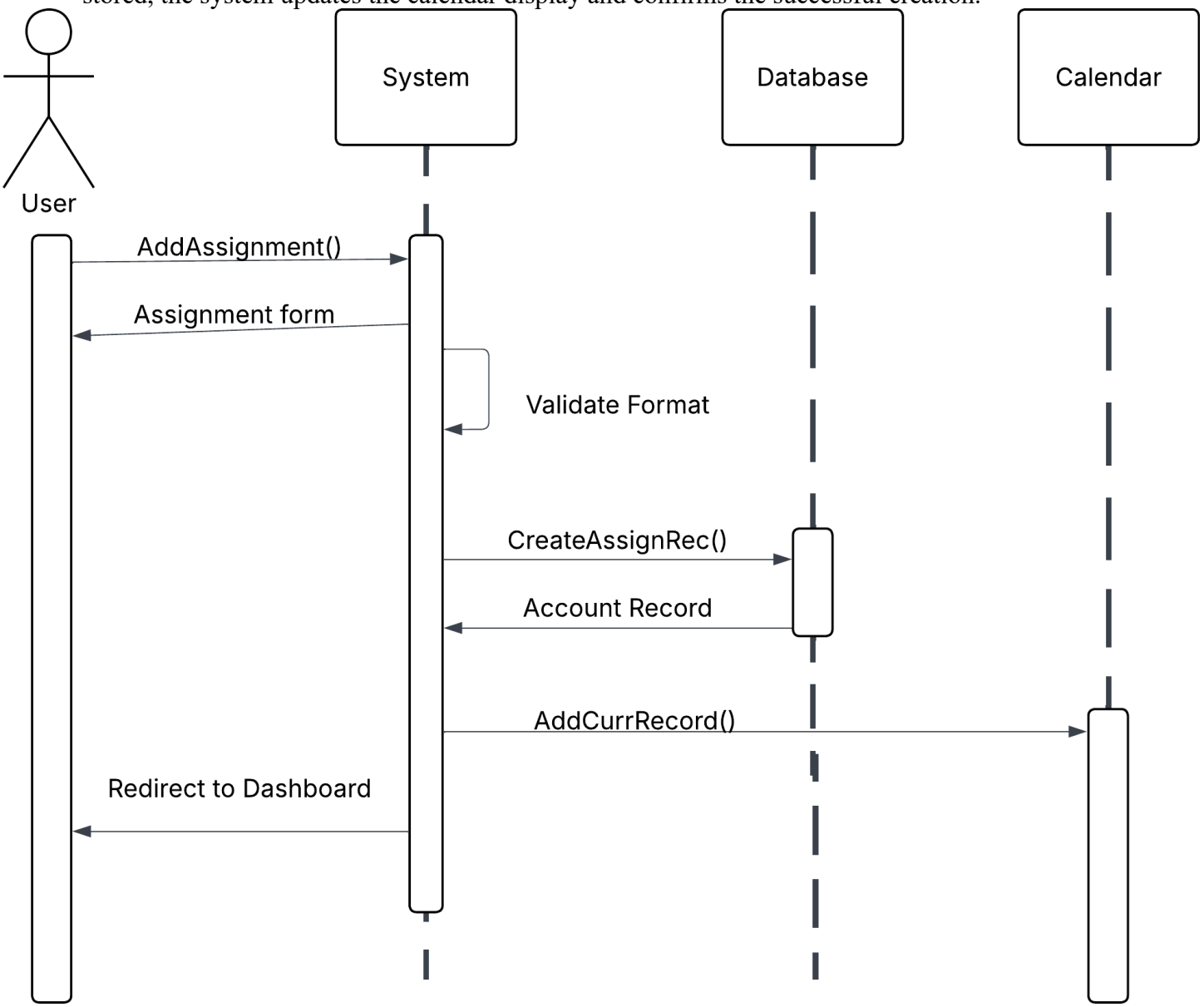

Figure 4 Data Flow Diagram Use Case 4

4.1.5 Use Case 5

The following diagram illustrates the process of creating a new project and assigning the project manager role. The process begins when the user selects to create a project and submits the project details. The system validates the input and verifies that the project name is unique within the database. The system then stores the project record and assigns the creator to the project manager role. The project is then added to the user's calendar, and the system displays a confirmation message along with returning the user to the dashboard.

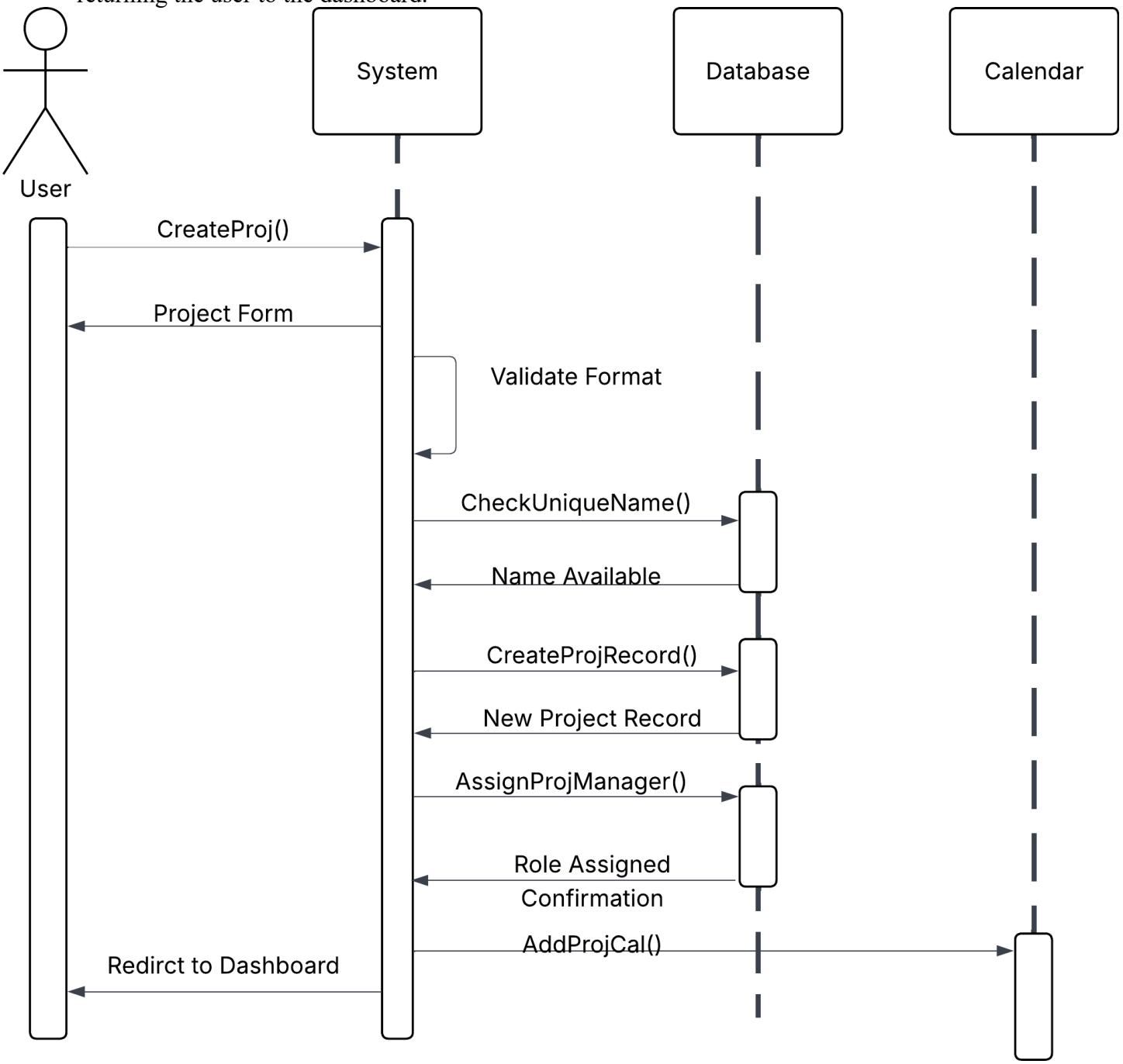

Figure 5 Data Flow Diagram Use Case 5

4.1.6 Use Case 6

The following diagram illustrates how a project manager assigns workers to a project phase. The process begins when the project manager selects the worker assignment option. The system retrieves a list of available users from the database and displays the worker selection interface. The project manager selects workers to assign and submits the selection. The system stores the assignment relationships and updates the users' calendars to include the project phase. The project manager is then redirected to the dashboard.

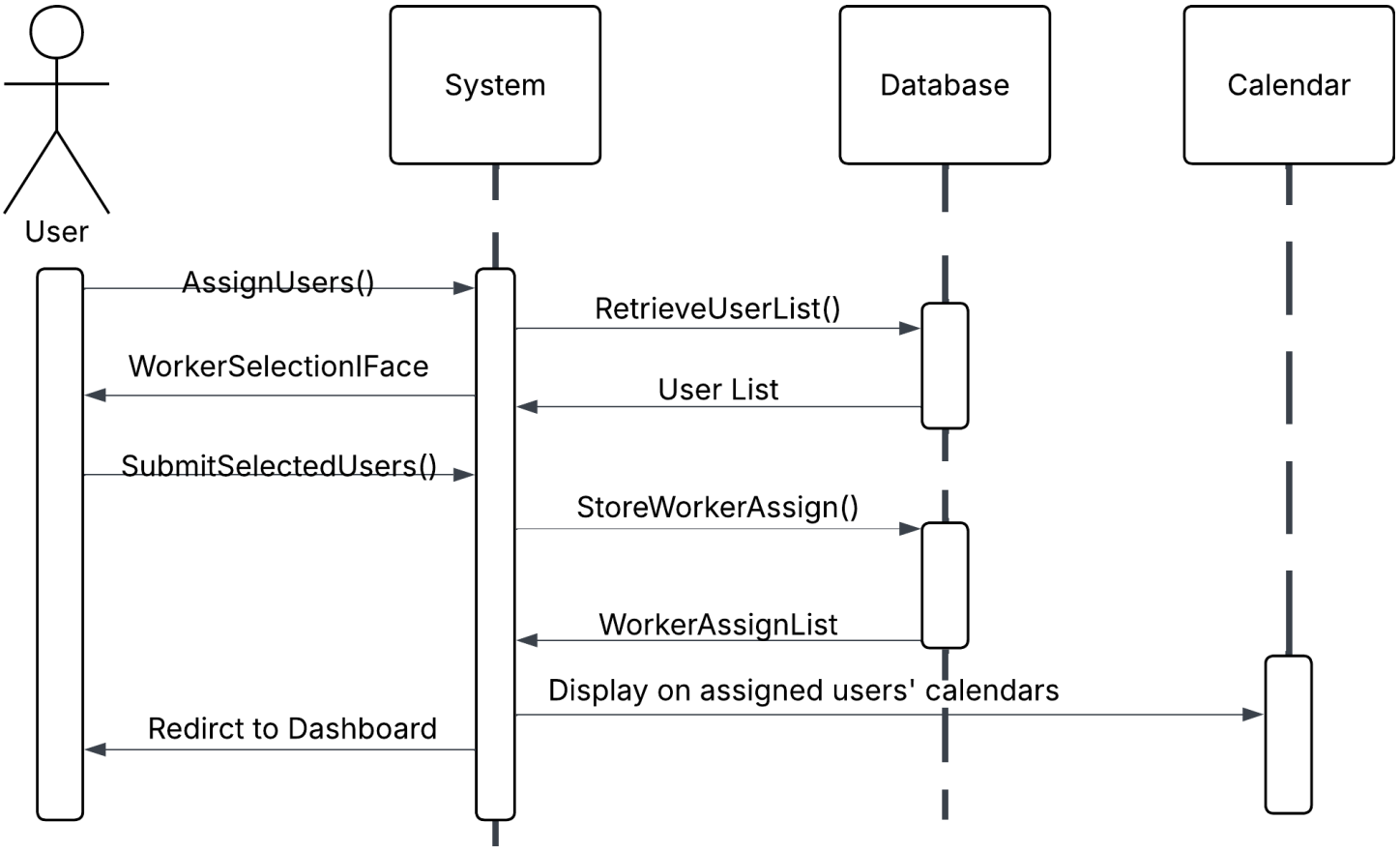

Figure 6 Data Flow Diagram Use Case 6

4.1.7 Use Case 7

The following diagram illustrates the process of creating and associating a new project phase. The process begins when the project manager selects the option to add a project phase. The system presents the phase creation interface, and the project manager submits the phase details. The system validates the input and stores the new phase associated with the project in the database. The system updates the calendar display to include the new phase and confirms successful creation.

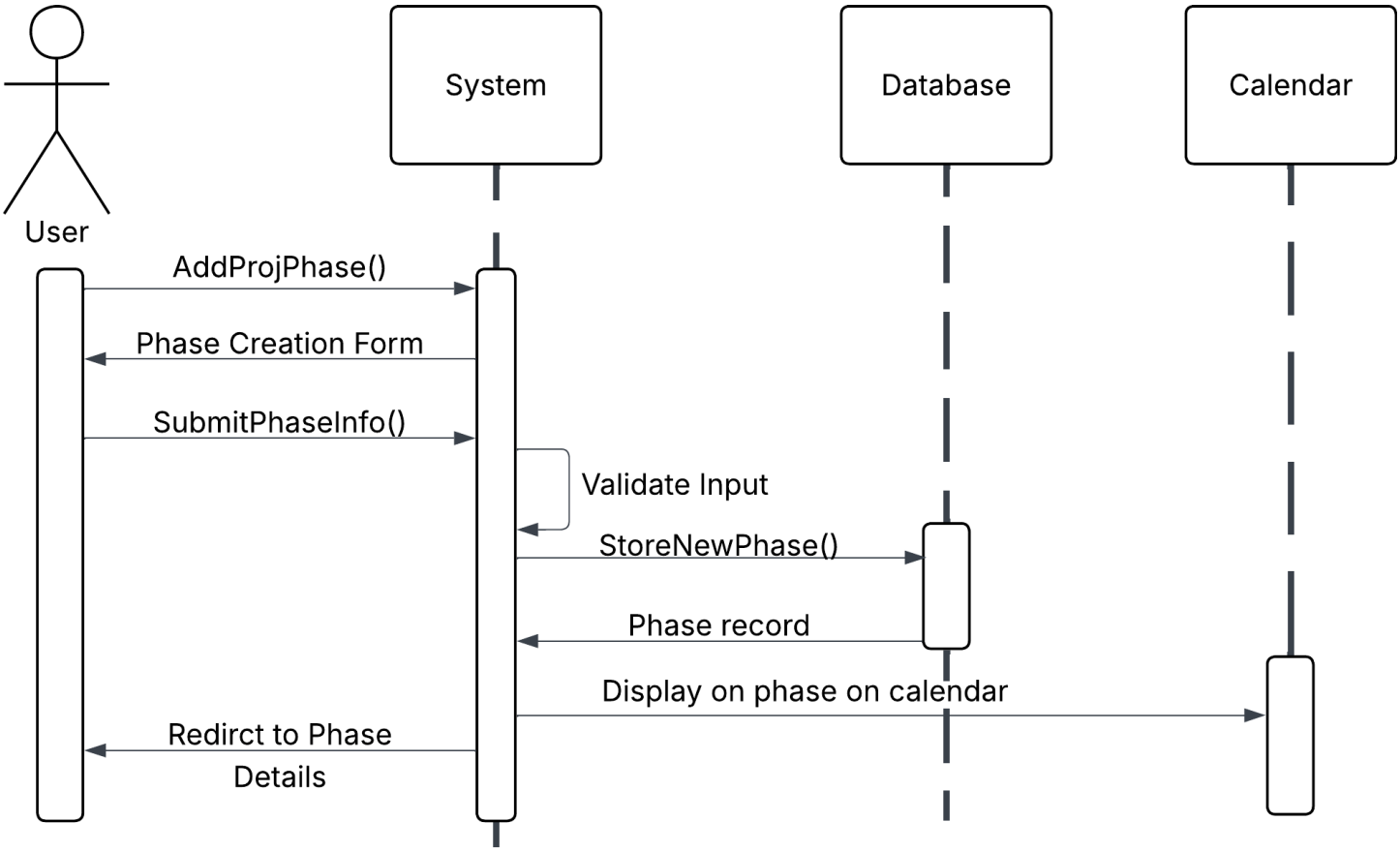

Figure 7 Data Flow Diagram Use Case 7

4.1.8 Use Case 8

The following diagram illustrates how project details are retrieved and displayed to the user. The process begins when the user selects a project from the dashboard or project list. The system retrieves the project information from the database and presents the project details interface to the user. No data is modified during this process.

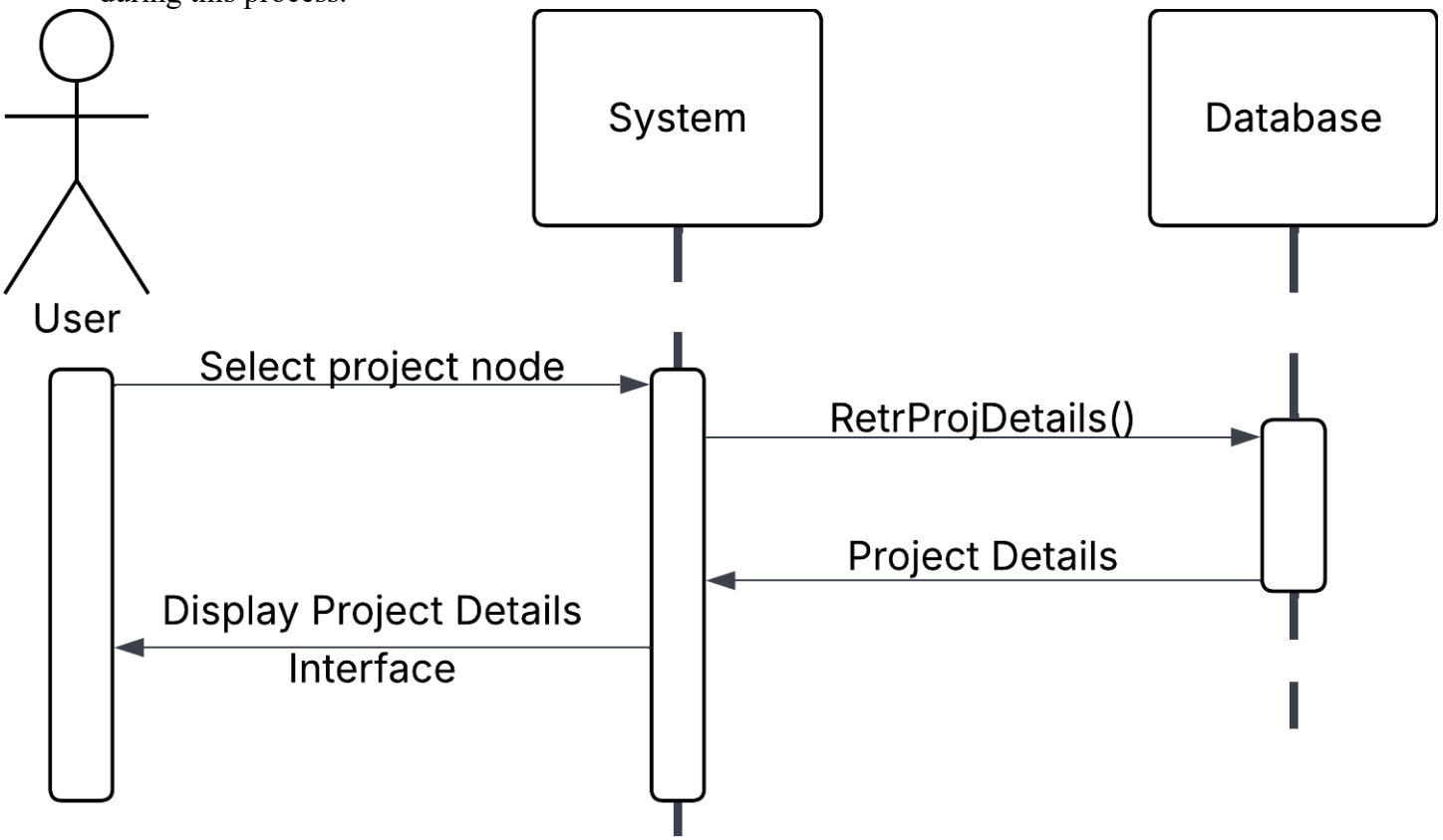

Figure 8 Data Flow Diagram Use Case 8

4.1.9 Use Case 9

The following diagram illustrates how the system retrieves and displays the list of projects associated with the user. The process begins when the user selects the project list option. The system retrieves all projects associated with the user from the database and orders them by due date. The system then displays the organized project list to the user. No data is modified during this process.

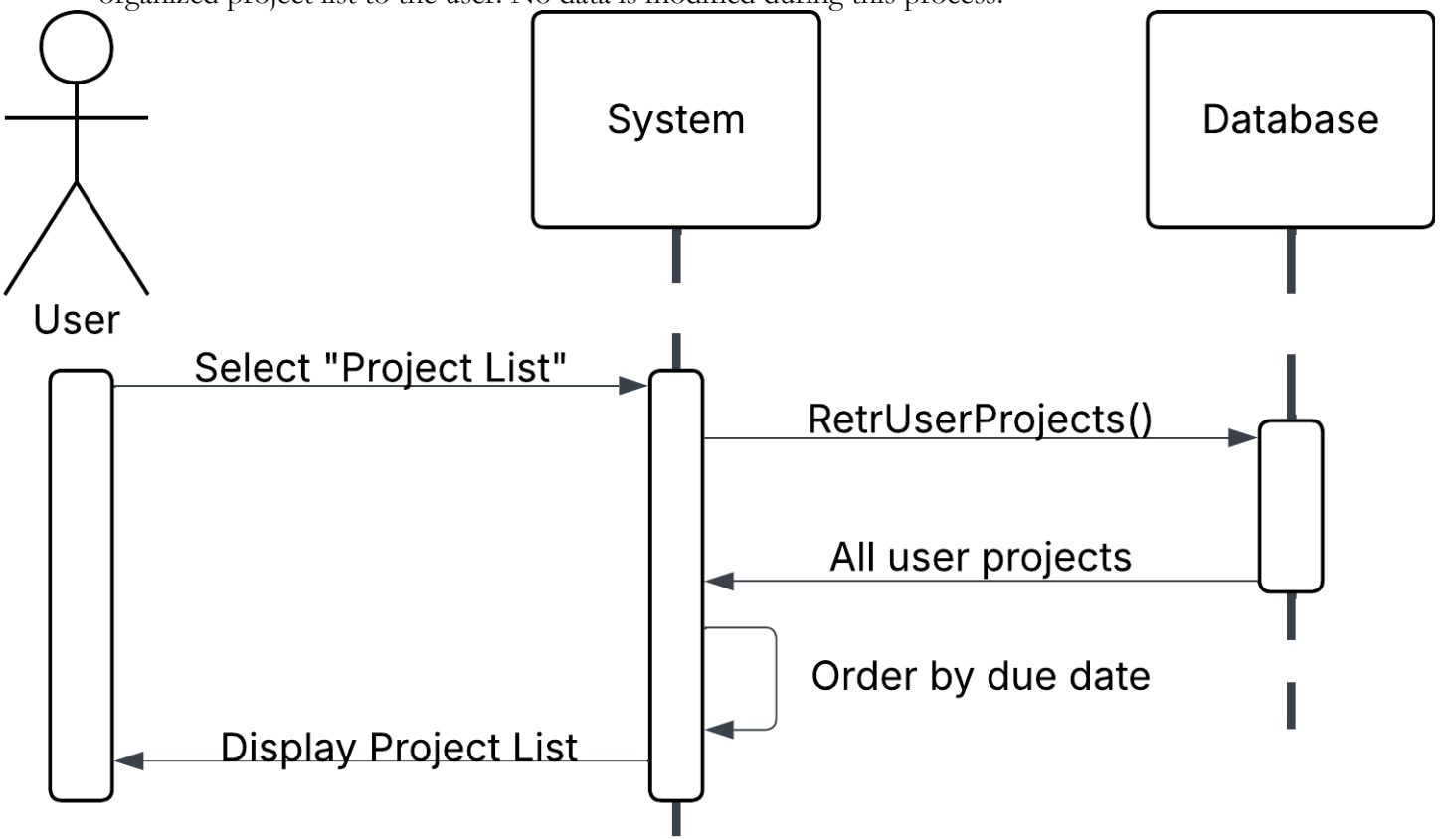

Figure 9 Data Flow Diagram Use Case 9

4.1.10 Use Case 10

The following diagram illustrates how a project manager archives a project. The process begins when the project manager selects the archive option. The system requests authentication confirmation and verifies the credentials in the database. Upon successful verification, the system updates the project status to archived and removes it from active calendar views. The system displays a confirmation message and returns the project manager to the project list.

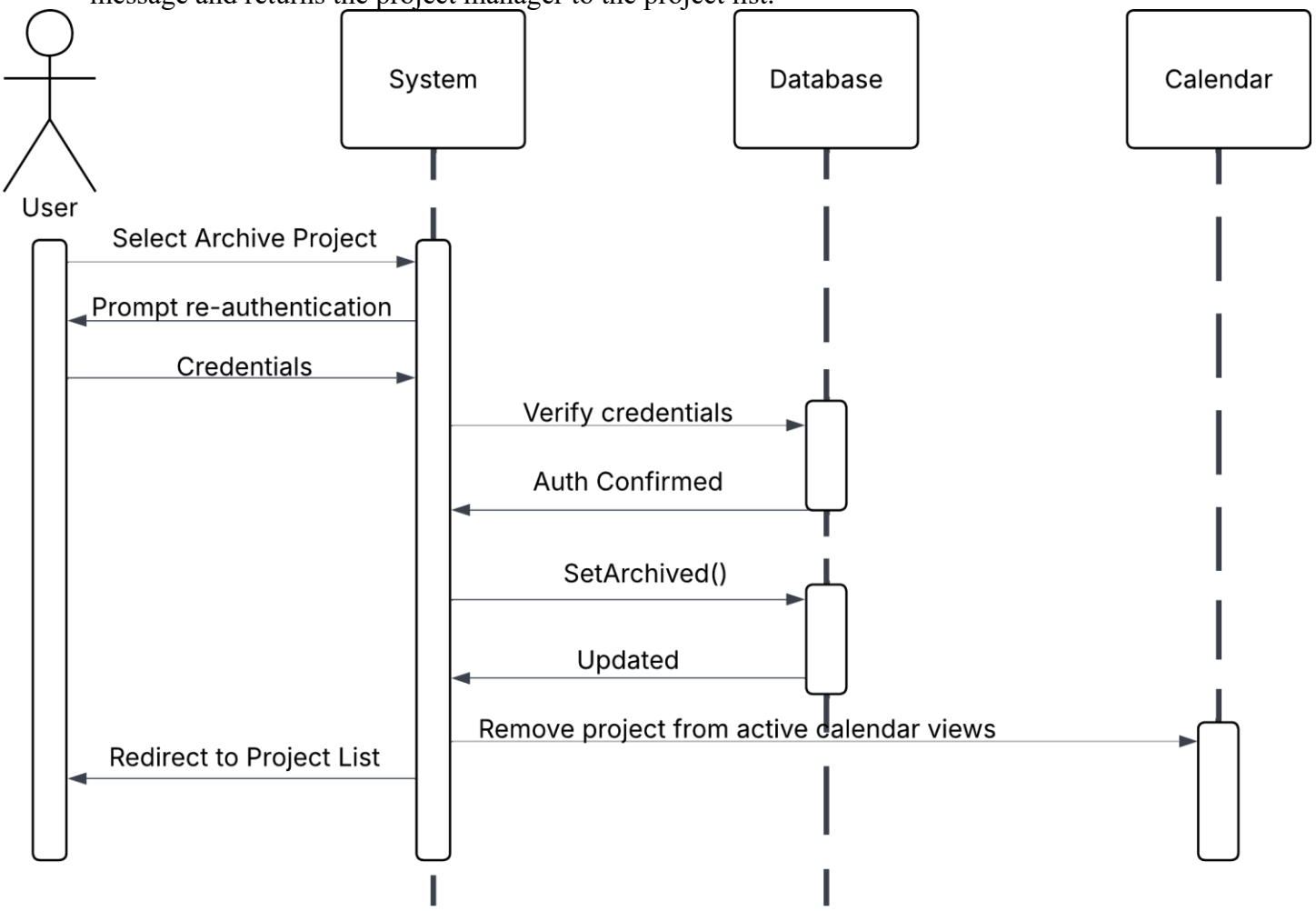

Figure 10 Data Flow Diagram Use Case 10

4.1.11 Use Case 11

The following diagram illustrates how a project manager modifies project information. The process begins when the project manager selects the modify option. The system requests authentication confirmation before allowing modification. After successful authentication, the system presents the modification interface, and the project manager submits updated project details. The system updates the project record in the database, refreshes the calendar display, and confirms the modification.

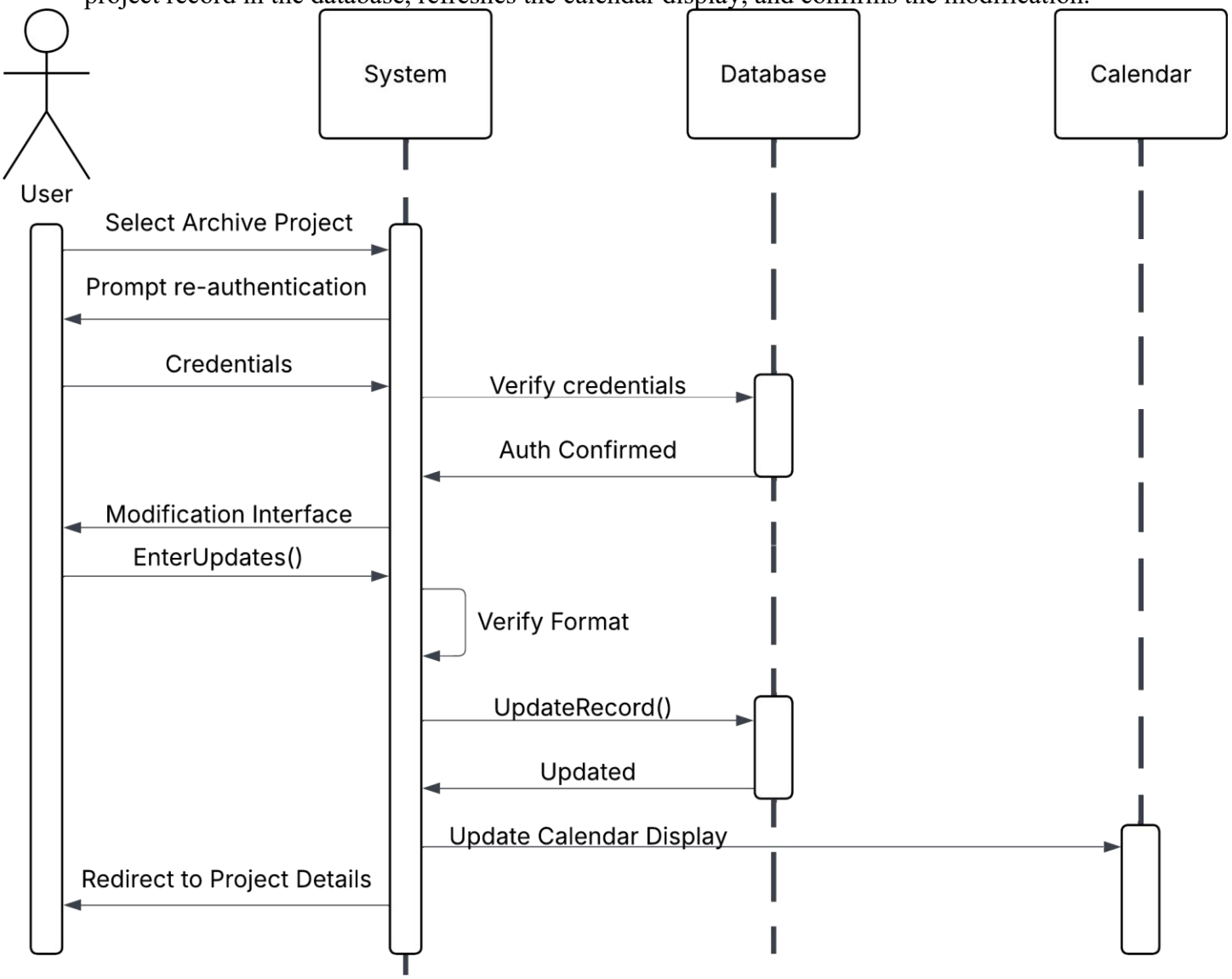

Figure 11 Data Flow Diagram Use Case 11

4.1.12 Use Case 12

The following diagram illustrates how dependencies between project phases are created and recorded. The process begins when the project manager selects the dependency creation option. The system retrieves and displays available project phases from the database. The project manager selects the source and dependent phases and submits. The system stores the dependency relationship in the database and updates the calendar to reflect the dependency. The system confirms the creation.

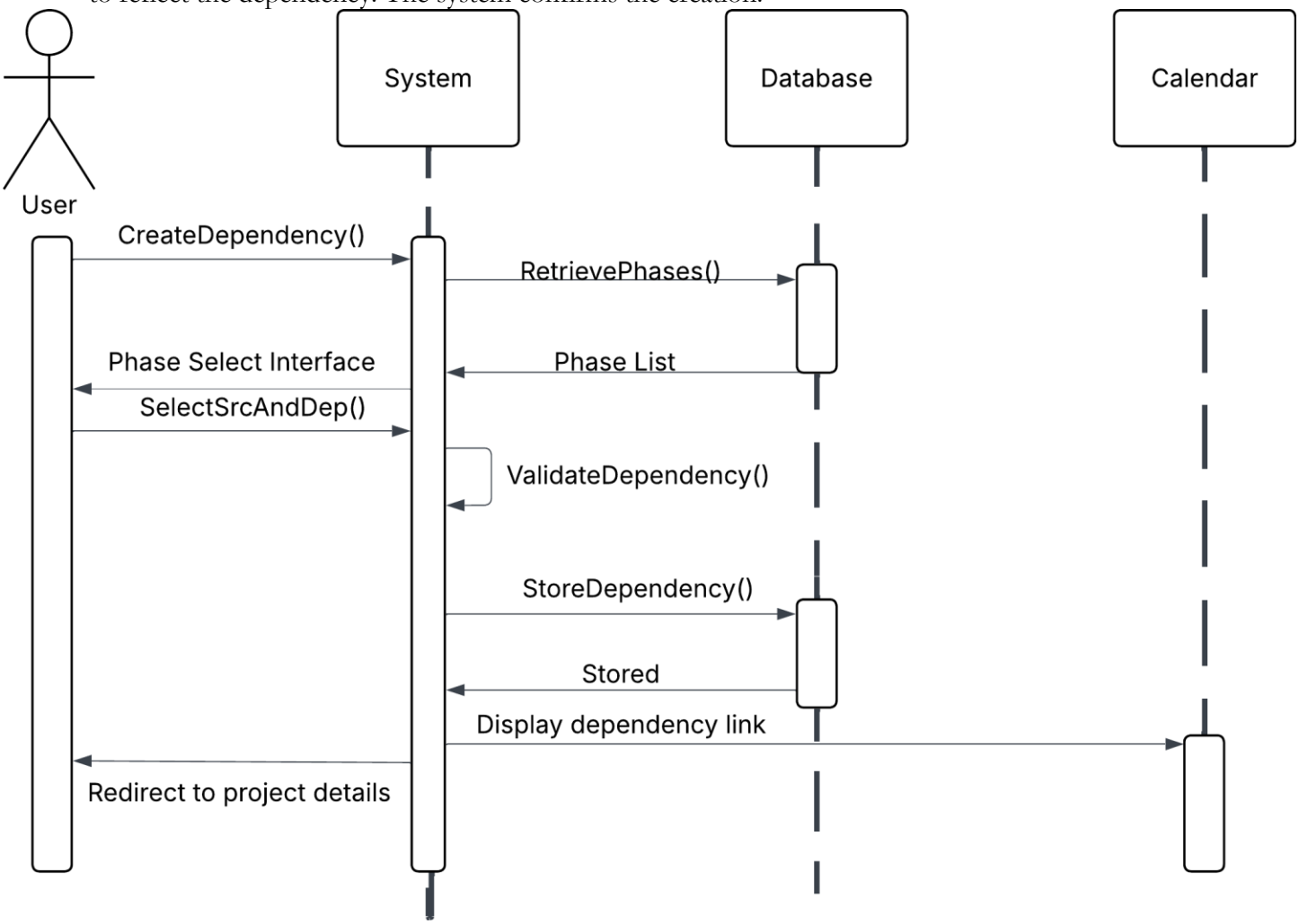

Figure 12 Data Flow Diagram Use Case 12

5. Change Management Process

Any future changes to the SRS document will be performed incrementally rather than large singular changes to prevent cascading effects to the rest of the document. Changes will be performed in one subsection at a time, and each section of the document must be re-examined to ensure any new change has not invalidated existing sections of the document. For any changes to the SRS, unilateral agreement from all project members is required. Changes which gain unilateral approval will then be implemented directly to the document by the project Team Lead.

References

A. Appendices

A.1 Appendix 1

A.2 Appendix 2